IT PROJECT MANAGEMENT — TUTORIAL

Cost & Scope Management

Learning Outcomes

- Decompose a project into a Work Breakdown Structure (WBS)
- Distinguish scope creep, gold plating, and legitimate change control
- Select the right cost estimating technique for a given situation
- Calculate and interpret Earned Value Management (EVM) metrics

EXERCISE 1

Build a Work Breakdown Structure

Scenario: Your team has been asked to deliver a 'Campus Food Ordering App' — a mobile app that lets students pre-order food from campus cafeterias and pick it up without queuing. The project must be delivered in one semester.

Step 1: Individually, list at least 8 deliverables or tasks you think are needed to build this app.

Step 2: Organize your list into a 3-level WBS — Level 0 (the project), Level 1 (major deliverables), Level 2 (work packages) — using the table below. Add rows if you need more.

Step 3: Compare your WBS with a partner. Discuss: is every Level 2 work package small enough to estimate, assign to one person, and track?

Level 0 (Project)

WBS Table — Level 1 Deliverables and Level 2 Work Packages

WBS ID	Level 1 — Deliverable	Level 2 — Work Package(s)
1.1		
1.2		
1.3		
1.4		
1.5		

Checkpoint: does every work package pass the '8/80 rule of thumb' — small enough to complete in roughly 8 to 80 hours of effort? If not, decompose it further.

EXERCISE 2

Scope Statement & Scope Creep vs. Gold Plating

Part A: For each situation below, decide whether it describes Scope Creep, Gold Plating, or a properly managed Change Request, and briefly justify your answer.

#	Situation	Your Answer + Justification
1	The client emails the developer directly asking for a 'quick extra filter option'; the developer adds it without telling the PM.	
2	A QA tester notices the checkout screen looks plain, so she adds animated transitions on her own initiative before release.	
3	The sponsor formally requests an extra payment gateway; the PM logs it, assesses impact on cost/schedule, and gets sponsor sign-off before building it.	
4	Over several sprints, small 'just one more field' requests pile up from different stakeholders with no one tracking the cumulative impact.	

Part B: Write a Mini Scope Statement

Using the Campus Food Ordering App scenario from Exercise 1, write one line each for the following components of a Project Scope Statement:

Product scope description:

One deliverable:

One exclusion (explicitly out of scope):

One acceptance criterion:

EXERCISE 3 ·

Cost Estimating Techniques

Part A: Match each technique to its correct description by writing the letter (A–D) in the blank column.

Technique	Answer (A–D)	Description Options
1. Analogous estimating		A. Estimate each work package individually, then sum them for a total
2. Parametric estimating		B. Use a statistical relationship, e.g. cost per line of code or per story point
3. Bottom-up estimating		C. Use the actual cost of a similar past project, scaled up or down
4. Three-point (PERT) estimating		D. Combine optimistic, most likely, and pessimistic values to account for uncertainty

Part B: Estimate Accuracy Ranges

Fill in the missing estimate type or accuracy range for each project stage:

Project Stage	Estimate Type	Typical Accuracy Range
Initiation		-25% to +75%
Planning	Budget Estimate	
Detailed design	Definitive Estimate	

EXERCISE 4

Earned Value Management (EVM) Calculations

Scenario: The 'Campus Food Ordering App' has a total approved budget (BAC) of LKR 800,000, spread evenly across an 8-month plan. You are reviewing status at the end of Month 4.

Given: Planned Value (PV) = LKR 400,000 | Earned Value (EV) = LKR 340,000 | Actual Cost (AC) = LKR 380,000

Show your working for each formula below, then write your final answer on the ruled line.

CV = EV – AC = _____

SV = EV – PV = _____

CPI = EV / AC = _____

SPI = EV / PV = _____

EAC = BAC / CPI = _____

ETC = EAC – AC = _____

VAC = BAC – EAC = _____

In one or two sentences, describe the overall health of this project (cost and schedule) and what you would recommend to the sponsor: